Multimedia Tools and Applications (2023) 82:26577–26593
https://doi.org/10.1007/s11042-023-14964-3

# A novel design of audio signals encryption with substitution permutation network based on the Genesio-Tesi chaotic system

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Received: 2 February 2022 / Revised: 14 February 2023 / Accepted: 22 February 2023 /
Published online: 9 March 2023
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**Abstract**
The concept of encryption is as old as the reality that confidential information has been interchanging among people. The art of transferring secret information between two parties or groups is known as cryptography. It contains the algorithms, key exchange protocols, and approaches to consistently and securely avoid unauthorized access to confidential information. Nowadays information is in digital form, therefore this factor necessitates some cryptographic algorithms for secure data transmission. In this work, we have designed a new approach for audio encryption to securely store and transfer audio signals. The suggested encryption algorithm is primarily based on Gensio-Tesi chaotic map. Substitution and permutation networks are engendered by the Gensio-Tesi chaotic system for the encryption of signals. The other important feature discussed here is the performance of offered algorithm which is ascertained by some extensive audio analysis. Experimental results reveal that the designed scheme has an essential level of security.

**Keywords** Audio encryption · Chaotic system · Spectrogram

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## 1 Introduction

Rapid expansions of computer technologies and extensive recognition of the internet influence the development of electronic data interchange and digital telecommunications. Subsequently, digital audio communication is utilized in routine activities such as commerce, banking, e-learning, politics, education, and the military. Therefore, an enormous amount of crucial audio records is exchanged daily over open and shared networks. As a result of the quick evolution of digital audio and data communications, the topic of supporting a high level of audio security becomes of the main importance. Cryptography provides security to digital data with robust encryption algorithms. Many traditional schemes have been developed for this purpose such as Data Encryption Standard (DES) [23], International Data Encryption Standard (IDEA) [5], Advanced Encryption Standard (AES) [2], Blowfish [19], Twofish, Serpent, etc. All the conventional schemes are utilized for data or image encryption and exhibit a complex structure for audio encryption. For this purpose, many audio encryption schemes have been developed based on chaotic systems to handle complex inputs such as audio. Chaotic systems are uncomplicated in design but generate signals of astonishing complexity, assisting numerous functions in cryptography. Chaotic maps have been utilized in recent times in cryptography for significant data encryption such as image, audio, and video data, as of their robust assets such as initial conditions sensitivity and variations in system control parameters, aperiodicity, and pseudo-randomness.

### 1.1 Literature review

A recent study in this field has revealed that the usage of chaotic systems for data encryption methods refers to elevated concentrations of security. Lin et al. [12] offered AES CFB (Advanced encryption standard cipher feedback) cryptosystem with SDKGs to encrypt the audio data. Arunkumar and Mukesh [4] utilized 2D zaslavsky map values for the key generation of the AES algorithm which was then structured for audio encryption. The key generation by chaotic map allows a large keyspace to repel brute force attacks. Authors in [18] suggested a new audio encryption design which is first compressed by Discrete Cosine Transform (DCT) to decrease the outstanding fluency. The compressed voice stream was shuffled by a hybrid chaotic system structured with Zaslavsky and Zigzag maps. A reference voice signal was created by a Hidden Markov Model (HMM) speech synthesizer for the substitution process and permuted utilizing a hyper-chaotic system. Shah et al. [20] developed an extensive data encryption scheme for image, audio, and video based on the Henon map and Mobius transformation. Abdelfatah [1] designed a new audio encryption approach with three phases. The first phase is a self-adaptive bit scrambling, the second is a dynamic DNA encoding of the permuted audio using multi-chaotic maps such as Sine, Chebyshev, and Logistic, and the third phase is two DNA algebraic operations “AND” and “XOR” utilizing a pseudorandom DNA sequence produced by alternative newly suggested multi chaotic HLG map that cartels Henon, Logistic, and Gaussian chaotic maps. Al-Hazaimeh [3] designed an innovative audio encryption model with innovative management of key and Jacobian elliptic maps. Hassan et al. [8] offered a digital audio encryption technique by arranging the audio files as a six-sided cubic model to scatter data. Encryption was performed by using Gingerbread and Hénon chaotic map. Farsana et al. [6] presented an audio scheme based on mixed chaotic keystreams and fast Hadamard transform. Shah et al. [21] proposed an innovative three-

dimensional chaotic map and utilize it in audio encryption with the process of confusion and diffusion. In [13] Liu et al. introduced an encryption technique based on multi-scroll chaotic systems that use confusion and diffusion. In [9] Hato and Shihab evaluated the Lorenz and Rossler chaotic system for voice signal encryption. More useful research is reported in [17] where Sathiyamurthi and Ramakrishnan exploited Bernoulli's chaotic map to build an encryption scheme. In [22] Tamimi and Abdalla offer another comparable strategy for an audio shuffle-encryption algorithm. Important research [14] shows that employing a single chaotic map is not necessarily a guarantee of maximum cryptographic security. The study examines how encryption schemes based on Arnold's Cat Map, Baker's Map, or a Two-Dimensional Logistic Chaotic Map can be compromised. Kordov [10] proposed a chaotic bit-level substitution and permutation-based audio structure. Lima and Silva presented a cosine number transform-based audio encryption algorithm in [11]. Several procedures for digital information encryption which utilize different shuffling methods during the encryption procedure were proposed in the literature, but all of the encryption schemes do not provide high security with low computational complexity [7, 15, 16]. Therefore, we aim for the construction of a secure audio encryption system to ensure confidentiality. Chaos is considered the most reliable source of pseudo-random number generators. Thus we have utilized a chaotic map to construct a highly secure cryptosystem with low computational complexity.

### 1.2 Research contribution

In this research, we have proposed a substitution permutation network-based audio encryption system based on the chaotic map. The main operations for the development of the encryption algorithms are shuffling and substitution by using the Gensio-Tesi map. The key contributions of this research are as follows:

1. The confusion mechanism is performed by the utilization of a chaotic substitution box generated from one component output of the Gensio-Tesi system.
2. The shuffling process is completed by applying a permutation array to the audio signals acquired from the substitution mechanism.
3. Furthermore, the ensuing confused audio is set to the random bit exclusive or operation. Consequently, all the enciphered audios will be affected by just a one-bit change in the audio sample.
4. An encrypted key is utilized to extract the initial conditions and control parameters; therefore, the system is key-sensitive. Integrity is ensured by adding parity.
5. Frequency, amplitude, spectrum, spectrogram, NIST, histogram, distance-based, and differential attack analysis are performed which show that the designed scheme is secure and can be employed in audio encryption.
6. The offered structure ensures the maximization of encryption speed and minimization of encryption time with robust security.

### 1.3 Paper organization

The rest of the manuscript is arranged as follows: Section 2 presents some basic concepts related to the Gensio-Tesi chaotic map with its stability and randomness analysis, the proposed

chaotic substitution box is listed in Section 3, and the structure of the designed work is defined in Section 4, in Section 5 we have discussed the experimental demonstration of the designed scheme and finally a conclusion is presented in the last section.

## 2 Some basic concepts

### 2.1 Genesio-Tesi chaotic system

In 1992 the Genesio–Tesi chaotic system, offered by Genesio and Tesi [7], is one of the epitomes of chaos as it describes many aspects of chaotic systems. It incorporates three simple ordinary differential equations and a square part that hinge on three real positive factors. The dynamic equation of the Genesio-Tesi chaotic system is formulated as:

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= z, \\ \dot{z} &= -cx - by - az + x^2, \end{aligned} \tag{1}$$

where $a$,$b$ and $c$ are positive real parameters with the condition $ab < c$. For the chaotic behavior in the system, we set the values of parameters. Initial conditions are $x_0 = 0.1$, $y_0 = 0.1$ and $z_0 = 0.1$ with parameters $a = 1$, $b = 3.03$, $c = 5.55$.

### 2.2 Stability analysis

The equilibrium state is denoted by $E(x, y, z)$ and analysis is performed by taking the values of state variables as $E_0(0, 0, 0)$. To determine an equilibrium state system (1) can be inscribed as

$$\begin{aligned} 0 &= y, \\ 0 &= z, \\ 0 &= -cx - by - az + x^2. \end{aligned} \tag{2}$$

The jacobian matrix of this system can be inscribed as

$$J(x, y, z) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5.55 + 2x & -3.03 & -1 \end{bmatrix} \tag{3}$$

For equilibrium points $E_0(0, 0, 0)$ the Jacobian matrix converts

$$J(0, 0, 0) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5.55 & -3.03 & -1 \end{bmatrix} \tag{4}$$

The eigenvalues for system (1) can be attained after solving the characteristic equation

$$|\lambda I - J| = 0 \tag{5}$$

After solving eq. (5) we get $\lambda^3 + \lambda^2 + 3.03\lambda + 5.55 = 0$, which gives three solutions $\lambda_1 = -1.482$, $\lambda_2 = 0.241 + 1.920\ i$, $\lambda_3 = 0.241 - 1.920\ i$. These eigenvalues demonstrate that the system has unstable spiral behavior which indicates the presence of chaos.

### 2.3 Randomness analysis

Chaotic systems produce an endless sequence of random bits. Some chaotic system exhibits supreme randomness in the subsequent statistics. Uncertainty in output data can be determined through various assessments. Here we are plotting the numerical statistics attained from the chaotic map along with several iterations *n*. Figure 1 presents the chaotic plot for the first 300 values along the *x, y,* and *z* directions.

Figure 2 shows the randomness plot for the first 3000 values in the *x, y,* and *z* directions. This visual presentation of data indicates the randomization in the output of a chaotic map.

## 3 Suggested chaotic substitution box (S-box)

In modern cryptology block cipher is one of the basic tools for building blocks. S-box architecture is of immense significance for assembling effective block encryption methods and is utilized in encryption structures. For assembling secure encryption approaches the structure of S-boxes with comprehensive cryptographic features is of ultimate importance. There exist many S-box generation ideas in literature from which chaotic systems are most powerful to enhance randomness. In this work, an S-box based on a chaotic system is designed. We have suggested a secure and ideal substitution box that exhibits the necessary scrambling effect against statistical attacks (Table 1).

## 4 Designed audio encryption model

In this segment, we have offered the structural design of the suggested audio encryption system using the Gensio-Tesi chaotic structure based on two main stages first one is substitution and the second is permutation. Some standard audio files are considered input data in the proposed encryption algorithm. The suggested audio encryption scheme contains the following strides:

Step 1: Set initial parameters for the Genesio-Tesi map.
Step 2: Iterate the chaotic map to get outputs of the x, y, and z components.
Step 3: Create an S-box from the x-component output of the Genesio-Tesi chaotic system.
Step 4: Generate a random sequence from the y-component output of the chaotic system.

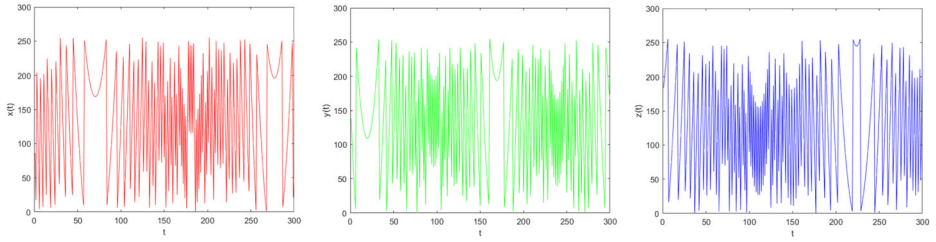



**Fig. 1** **a** Plot of Genesio-Tesi map for 300 values along *x* and *t*; **b** along *y* and *t*; **c** along *z* and *t*

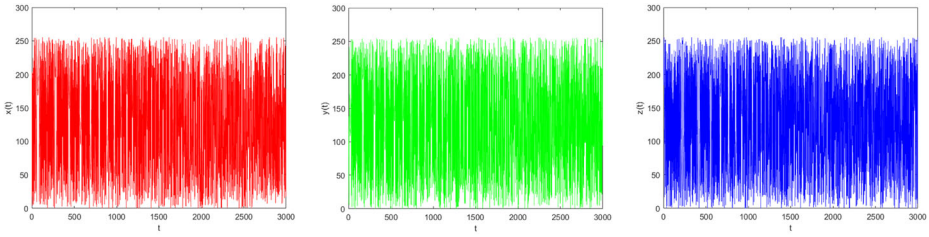



**Fig. 2** **a** Plot of Genesio-Tesi map for 3000 values along $x$ and $t$; **b** along $y$ and $t$; **c** along $z$ and $t$

Step 5: Design an arbitrary permutation array from the z-component output of a chaotic system.
Step 6: Read an audio file as input data in the encryption algorithm.
Step 7: The plain data of the audio file ranges from [−1,1] and it is converted in the form [0,255] by applying some transformation.
Step 8: Apply the S-box created in 3 on the plain data file obtained in step 7.
Step 9: Random array generated in step 4 is Xor-ed with the results from step 8.
Step 10: The permutation array obtained from step 3 is applied to the output of step 9.

The output of step 10 is the complied in audio format. The design of the suggested audio encryption algorithm is depicted in Fig. 3. Each stage of encryption is visibly characterized in the flowchart of the designed algorithm.

The decryption of the suggested encryption algorithm requires all the encryption steps in a reverse manner. The main decryption steps include:

1. Implementation of the inverse permutation of the encrypted audio data.
2. Bitwise Xor of the key and the output of step 1.
3. Application of the inverse S-box transformation on data obtained from step 2.
4. The obtained results are then compiled as decrypted audio data.

**Table 1** Proposed chaotic substitution box

| 61 | 58 | 136 | 157 | 204 | 162 | 7 | 147 | 236 | 118 | 72 | 135 | 194 | 1 | 89 | 80 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 166 | 37 | 240 | 174 | 85 | 67 | 217 | 234 | 91 | 105 | 42 | 104 | 43 | 228 | 25 | 47 |
| 22 | 23 | 96 | 199 | 255 | 8 | 181 | 197 | 62 | 133 | 98 | 77 | 93 | 31 | 79 | 251 |
| 142 | 17 | 215 | 233 | 224 | 241 | 141 | 151 | 44 | 127 | 191 | 216 | 73 | 238 | 163 | 9 |
| 13 | 18 | 110 | 68 | 203 | 243 | 161 | 143 | 49 | 6 | 38 | 138 | 121 | 154 | 65 | 226 |
| 148 | 28 | 5 | 126 | 189 | 83 | 190 | 156 | 71 | 183 | 53 | 139 | 116 | 178 | 124 | 134 |
| 35 | 45 | 165 | 193 | 201 | 130 | 229 | 177 | 2 | 159 | 213 | 187 | 211 | 220 | 12 | 123 |
| 185 | 70 | 76 | 29 | 34 | 250 | 20 | 207 | 75 | 173 | 128 | 27 | 63 | 111 | 230 | 64 |
| 87 | 103 | 180 | 137 | 81 | 150 | 78 | 247 | 97 | 244 | 54 | 92 | 210 | 170 | 179 | 14 |
| 252 | 144 | 115 | 253 | 99 | 242 | 221 | 101 | 158 | 40 | 112 | 100 | 152 | 30 | 245 | 232 |
| 169 | 192 | 59 | 122 | 32 | 196 | 50 | 182 | 200 | 176 | 106 | 223 | 212 | 206 | 46 | 95 |
| 94 | 248 | 11 | 254 | 0 | 60 | 145 | 153 | 149 | 4 | 131 | 235 | 237 | 184 | 84 | 167 |
| 26 | 57 | 227 | 140 | 3 | 188 | 249 | 164 | 108 | 205 | 168 | 120 | 117 | 186 | 39 | 15 |
| 222 | 129 | 195 | 33 | 146 | 69 | 107 | 56 | 82 | 66 | 52 | 55 | 132 | 172 | 246 | 219 |
| 125 | 209 | 171 | 102 | 114 | 24 | 231 | 214 | 16 | 225 | 208 | 74 | 160 | 10 | 218 | 19 |
| 88 | 41 | 155 | 21 | 36 | 198 | 51 | 48 | 90 | 113 | 119 | 109 | 202 | 239 | 175 | 86 |

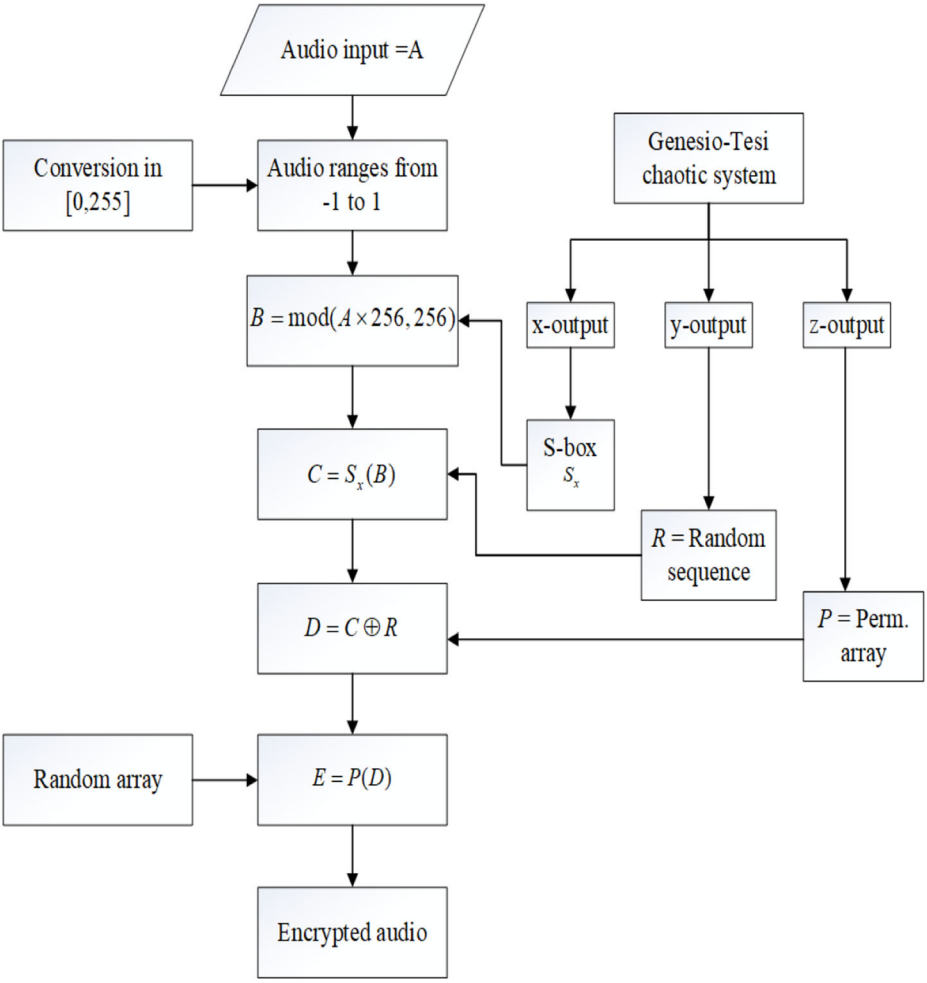



**Fig. 3** Design of offered audio encryption algorithm

## 5 Algorithm performance analysis

Audio analysis implies the lineage of meaning and information from audio signals for storage, synthesis, analysis, classification, retrieval, etc. The detection channels and clarification ways fluctuate; as audio assessment can have related to the human ear or it may refer to the utilization of equipment such as an audio analyzer. An ideal encryption scheme must have to be resiliently opposed to all current attacks. A competent encryption algorithm breaks all the connections between original and encrypted data due to which attackers are unable to get any meaningful link between plaintext and ciphertext. In this section of the article, we have implemented some analyses to test the competence of our suggested audio encryption algorithm which are frequency, amplitude, spectrum, spectrogram, NIST (National Institute of Standards and Technology), histogram, and distance-based robustness analysis.

### 5.1 Frequency analysis

The plot of frequency components is the frequency-domain representation of the signal. Any meaningful audio exhibits a frequency plot with a highly fluctuating curve but when the audio is encrypted its frequency plot displays uniform behavior. The uniformity in the encrypted audio refers to the loss of possible attack ways. Figure 4 indicates the frequency plot of the original and encrypted magic tune with a size of 106/kB. From the visual analysis of Fig. 4, we can remark that encrypted audio exhibits uniform frequency which increases the security of the suggested audio encryption algorithm.

### 5.2 Amplitude analysis

Amplitude is the displacement or fluctuation of a wave from its median value. To examine the amplitude of a periodic variable, change over a single period is measured. Amplitude plots of lion roar audio with size 50/kB are presented in Fig. 5. From Fig. 5 (a) we can examine the fluctuation of the amplitude of original audio data and when the same data is encrypted then it shows uniform and almost periodic behavior which shows maximum loss of meaningful data, as depicted in Fig. 5 (b).

### 5.3 Spectrum analysis

The dynamics of the audio signals are analyzed by a spectrum diagram. Variation of spectral contents depends upon time. Spectrum analysis is performed by using a spectrum analyzer, which measures the frequency versus magnitude of an input signal. In most widespread spectrum analyzers input signal is electrical however, spectral symphonies of other types of signals, such as optical light waves and acoustic pressure waves can be deemed beyond the utilization of a suitable transducer. For other types of signals, spectrum analyzers are also available, for instance, optical spectrum analyzers that utilize direct optical methods such as a monochromator to get observations. Spectrum analysis of electrical signals observes power, harmonics, bandwidth, dominant frequency, and other spectral aspects that are not simply measurable in time realm waveforms. In the modeling of many electronic gadgets such as wireless transmitters, all these parameters are helpful.

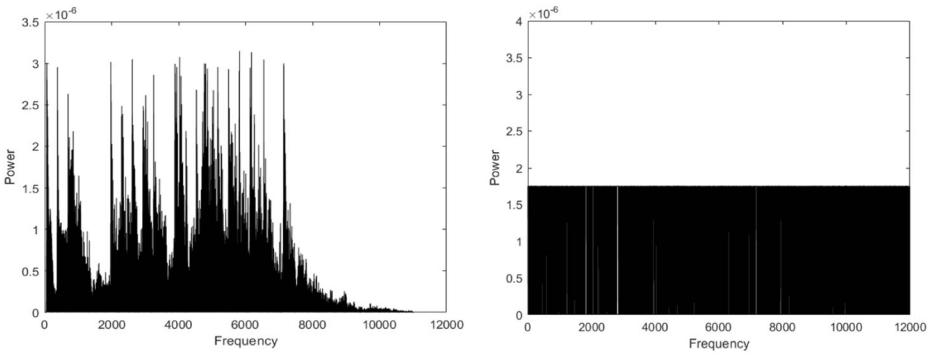



**Fig. 4** Frequency analysis of magic (**a**) original audio (**b**) encrypted audio

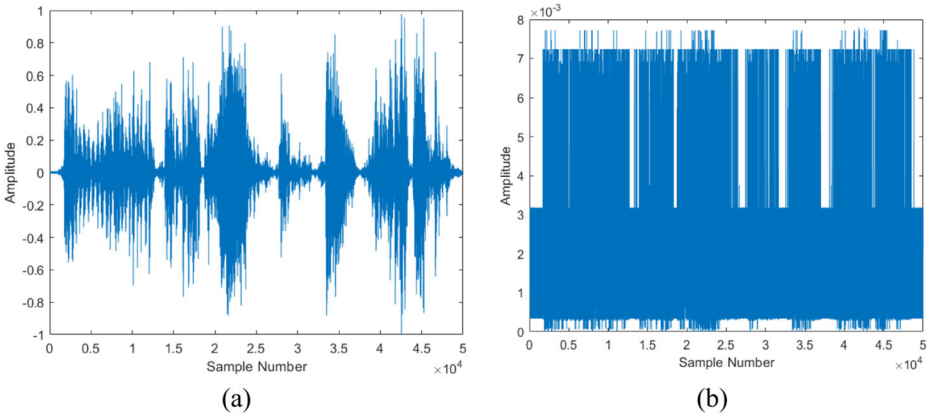



**Fig. 5** Amplitude analysis of lion roar (**a**) original audio (**b**) encrypted audio

The display of a spectrum analyzer has a horizontal axis as frequency and a vertical axis as amplitude. Figure 6 (a) and (b) offer the spectrum diagram of the original and enciphered audio signal respectively. From the figure, we can examine that the display between amplitude and frequency becomes almost uniform which indicates the loss of connection between the original and cipher data.

### 5.4 Spectrogram of signal

The visual representation of the variation of spectrum frequency concerning time is known as a spectrogram. The audio signal spectrogram is sometimes also referred to as voicegrams, voiceprints, or sonograph. Data representation in the 3D plot may also be called waterfalls. Spectrograms are extensively utilized in the fields of sonar, music, radar, seismology speech processing, and others. Figure 7 (a) and (b) present the spectrogram plot for original and encrypted audio data respectively. We can visualize from the figures that the spectrogram of encrypted data possesses almost the same frequency concerning the time as compared to the original data sonograph which signifies the efficiency of the encryption algorithm.

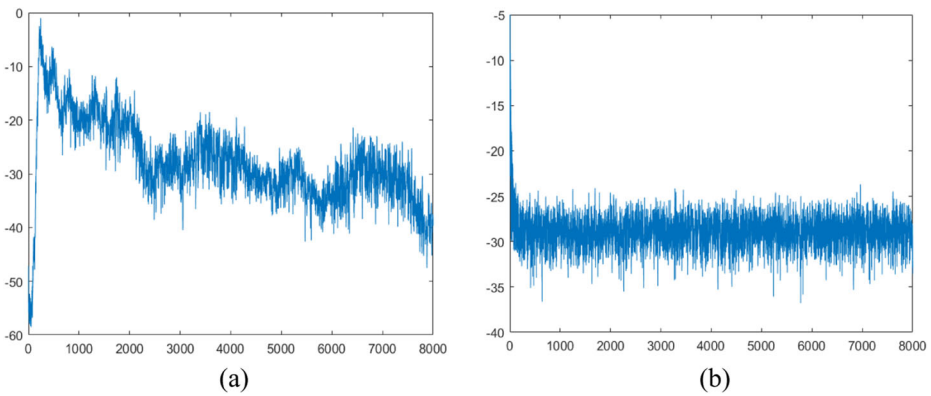



**Fig. 6** Loin roar spectrum analysis of (**a**) original signal; (**b**) enciphered signal

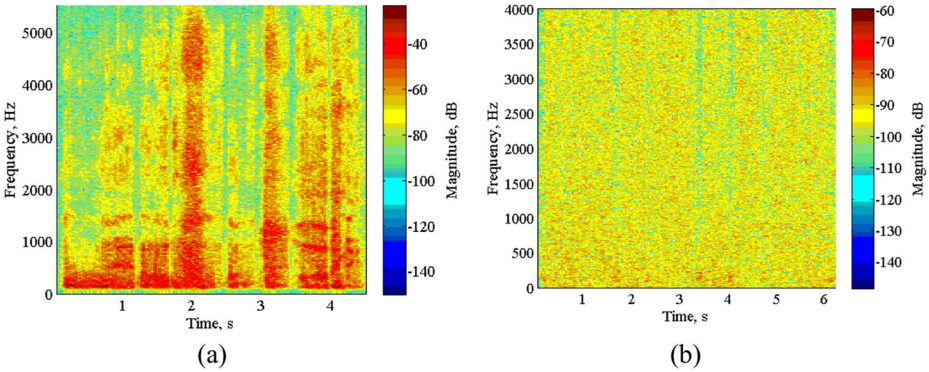



**Fig. 7** Loin roar spectrogram analysis of (**a**) original audio; (**b**) encrypted audio

## 5.5 Randomness analysis

NIST statistical test suite was offered by the National Institute of Standards and Technology which performs the randomness testing of data. Highly random data is utilized in many fields involving cryptographic, modeling, and simulation applications. The major focus of this assessment is to inspect the non-randomness of data. The NIST software performs 17 tests on a given binary sequence to determine the level of randomness. The actual assessment is carried out by segmenting the 1,000,000,000-bit sequence into 1000 subsequences, each of which has 1,000,000 bits. Some NIST tests are listed in Table 2 which demonstrates that encrypted audio data is highly randomized and secure against all possible randomness-based attacks. P-values in the [0,1) range indicate that all NIST tests have been passed.

**Table 2** NIST test outcomes for lion roar encrypted audio

| Test Name | | *p* values for encrypted audio signal | Results |
|---|---|---|---|
| Frequency | | 0.63526 | Success |
| Block-frequency | | 0.81819 | Success |
| Runs (M = 10,000) | | 0.34457 | Success |
| Long runs of one | | 0.03575 | Success |
| Rank | | 0.29191 | Success |
| Spectral DFT | | 0.66336 | Success |
| No overlapping templates | | 0.89638 | Success |
| Overlapping templates | | 0.85988 | Success |
| Universal | | 0.99901 | Success |
| Serial | *p* values 1 | 0.79284 | Success |
| Serial | *p* values 2 | 0.95563 | Success |
| Approximate entropy | | 0.98239 | Success |
| Cumulative sums forward | | 0.20988 | Success |
| Cumulative sums reverse | | 0.10469 | Success |
| Random-excursion | | 0.99551 | Success |
| Random-excursion variant | | 0.99740 | Success |
| Linear complexity | | 0.41033 | Success |

### 5.6 Histogram analysis

The estimation probability distribution of some variables is depicted by a histogram. The histogram is constructed after dividing the possible range values into small intervals known as bins. The bins are often of equal width (not necessarily) and must be contiguous to each other. The height of each bin is determined by how many numbers of values will fall in each bin and the number of cases in every bin is proportional to the height. An audio histogram is also utilized to present "relative" frequencies. Figure 8(a) and (b) display histograms of original and encrypted audio. Figure 8 (a) indicates that there are sharp curves in the original data histogram but after encryption curves become smooth as shown in Fig. 8 (b).

### 5.7 Similarities measures

Normalized cross-correlation and structure content are used to approximate the similarities between two signals. There exist some basic measures to analyze the analogist among signals. We have used consistent structure content and correlation with the completion objective to demonstrate the discrepancies between original and enciphered audio.

#### 5.7.1 Normalized cross-correlation

NCC (Normalized cross-correlation) is normally applied as a metric. NCC is used to measure the intensity of the difference between original and encrypted audio. The primary favorable position of the common cross-correlation and normalized cross-correlation is that it is sensitive to straight alterations in the adequacy of radiance in two examined audios. The range of normalized cross-correlation is from −1 and 1. Arithmetically normalized cross-correlation is formulated as:

$$NCC = \frac{\sum_{x=1}^{X} O_x \times E_x}{\sum_{x=1}^{X} {E_x}^2}, \tag{6}$$

where $X$ is the length of original audio $O$ and encipher audio $E$ respectively.

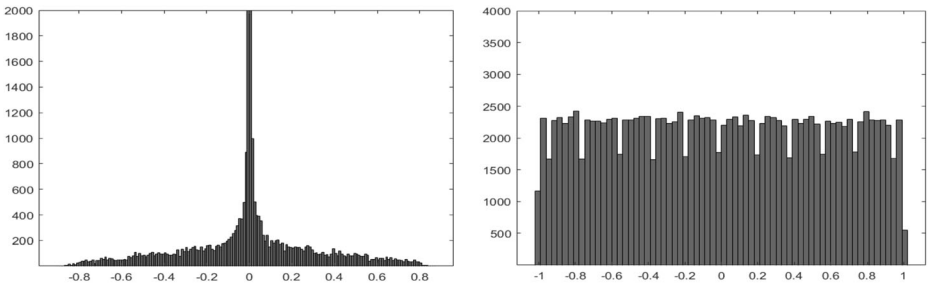



**Fig. 8** Histogram analysis of bird whistle audio (**a**) original signal; (**b**) enciphered signal

### 5.7.2 Structural content

Structural content analyzes the mutual weight of an original signal to that of an encrypted one. It is, hence, a global metric. This method is also known as SC and a surge in the valuation of SC from unity refers that the encrypted signal exhibiting low security against attacks. The calculated formulation of structural content is defined as:

$$SC = \frac{\sum_{x=1}^{X}(O(x,k))^2}{\sum_{x=1}^{X}(E(x,k))^2} \tag{7}$$

With the utilization of original and encrypted audio, structural content is measured and listed in Table 3. From examined results, we can see that the SC values are far away from unity and highly disordered in encrypted data.

### 5.8 Peak signal-to-noise ratio and mean square error analysis

Mean square error also referred to as MSE process the plain and enciphered audios to assess the level of encryption. MSE can be mathematically written as:

$$MSE = \frac{\sum_{x=1}^{X}(O_x - E_x)^2}{X}, \tag{8}$$

where $O_x$ and $E_x$ are the original and encrypted values of the signal at $x^{th}$ position respectively. The encrypted audio signal quality is assessed by using the peak signal-to-noise ratio (PSNR), which is characterized mathematically by

$$\mathrm{PSNR} = 20\log_{10}\left[\frac{I_{\max}}{\sqrt{MSE}}\right], \tag{9}$$

where $I_{\max}$ is the highest signal estimation of audio data. The outcomes of MSE and PSNR are represented in Table 4. A brief comparison of the proposed algorithm with the already published scheme is also enumerated in Table 4.

**Table 3** Similarity measures of suggested audio encryption technique

| Image | Audio Type | Similarity Measures | |
|---|---|---|---|
| | | SC | NCC |
| Lion Roar | Wav | 0.0247 | 0.0003 |
| Water Drops | Mp3 | 0.0039 | 0.0014 |
| Bird Whistle | Mp3 | 0.0494 | −0.003 |
| Magic | Mp3 | 0.0384 | −0.001 |
| Lazer | Mp3 | 0.8323 | 0.0090 |
| Jingle Bell | Mp3 | 0.0158 | 0.0588 |
| Car Lock | Mp3 | 0.0103 | 0.0495 |
| Hello | Wav | 0.0086 | −0.003 |
| Doorbell | Mp3 | 0.0032 | −0.062 |

**Table 4** Quality evaluation-based measures of suggested encryption technique

| Audio Name | Audio Size | Quality Evaluation Measures | |
|---|---|---|---|
| | | MSE | PSNR |
| Lion Roar | 50/kB | 0.0270 | 63.814 |
| Water Drops | 65/kB | 0.3342 | 52.891 |
| Bird Whistle | 31/kB | 0.3493 | 52.699 |
| Magic | 106/kB | 1.0343 | 47.984 |
| Lazer | 18/kB | 0.6057 | 50.308 |
| Jingle Bell | 111/kB | 1.0100 | 48.087 |
| Car Lock | 18/kB | 1.0054 | 48.107 |
| Hello | 170/kB | 1.0045 | 48.111 |
| Doorbell | 133/kB | 0.9996 | 48.133 |
| Ref. [20] | – | 0.3263 | 10.770 |
| Ref. [17] | – | – | 62.31 |

### 5.9 Robustness analysis

To examine the arbitrariness of the encryption algorithm, diffusion is a significant factor that must be approximated. Two traditional processes are employed known as UACI and NSCR also labeled as uniform average changing intensity number of signals change rate to examine the impact of one signal variation in the original and the enciphered audio data. NSCR essences on the number of signals which alter the assessment in differential assaults although the UACI focuses on the AD among two corresponding encrypted signals. Consider enciphered signals previously and after that one signal change in original audio data are $C_1$ and $C_2$ respectively. The value of signals at grid *(x)* in $E_1$ and $E_2$are signified as $E_1(x)$and $E_2(x)$ and $D$ a bipolar array is depicted by Eq. (12). The UACI and NSCR can be expressed by the following formula:

$$\mathrm{NSCR} = \frac{\sum_x D(x)}{X} \times 100\%, \tag{10}$$

where

$$D(x) = \begin{cases} 0, & E_1(x) = E_2(x) \\ 1, & E_1(x) \neq E_2(x) \end{cases} \tag{11}$$

$$UACI = \frac{1}{X} \sum_{x=0}^{X-1} \left| \frac{E_1(x) - E_2(x)}{255} \right| \times 100\%, \tag{12}$$

Table 5 represents the assessments of, UACI (≈33%) and NSCR (>99%) for some encrypted audio signals. Comparative analysis is also presented in Table 5. Investigational impacts explain the approximate divergence and probabilities of UACI and NSCR are close to the theoretic results. Hence our suggested encryption scheme is robust despite differential assaults.

**Table 5** Differential attack analysis of suggested encryption technique

| Audio Name | NSCR | UACI |
|---|---|---|
| Lion Roar | 99.99 | 33.73 |
| Water Drops | 99.92 | 33.41 |
| Bird Whistle | 99.90 | 33.60 |
| Magic | 99.96 | 33.45 |
| Lazer | 99.93 | 33.69 |
| Jingle Bell | 99.90 | 33.52 |
| Car Lock | 99.98 | 33.45 |
| Hello | 99.93 | 33.89 |
| Doorbell | 99.97 | 33.82 |
| Ref. [20] | 99.98 | 30.24 |
| Ref. [17] | 99.99 | 33.31 |

### 5.10 Execution time analysis

We constructed the encryption structure in Matlab 2019b while it was running on a personal computer with Windows 10, and a 64-bit operating system. This allowed us to determine how long the encryption scheme would take to execute. The personal computer has a processor that is an Intel (R) Core (TM) i7-5600U CPU running at 2.60 GHz and 16 gigabytes of RAM. The amount of time required to decrypt various audio files, each of which contains a unique set of characters, is detailed in Table 6.

### 5.11 DIEHARD-test

The DIEHARD program offers 19 randomness tests, which we repeated over the identical 1,000,000,000-bit bitstream given by our PRG. George Marsaglia worked on the DIEHARDS test for several years before putting it out in 1995 on a CD-ROM of random numbers. For estimated p-values, the allowable range for passing the single tests is [0, 1). Table 2 summarizes the findings of all testing. The results are also compared with Ref. [10] to check the accuracy of the offered scheme. The comparative results indicate the robustness of the proposed encryption model (Table 7).

**Table 6** Execution time analysis for the proposed audio encryption technique

| Image | Audio Type | Audio size | Execution time (s) | |
|---|---|---|---|---|
| | | | Proposed | Ref. [20] |
| Lion Roar | Wav | 50/kB | 0.41 | 1.230 |
| Water Drops | Mp3 | 65/kB | 0.97 | – |
| Bird Whistle | Mp3 | 31/kB | 0.33 | 0.612 |
| Magic | Mp3 | 106/kB | 1.60 | – |
| Lazer | Mp3 | 18/kB | 0.29 | – |
| Jingle Bell | Mp3 | 111/kB | 2.01 | 2.091 |
| Car Lock | Mp3 | 18/kB | 0.28 | – |
| Hello | Wav | 170/kB | 2.93 | – |
| Doorbell | Mp3 | 133/kB | 2.66 | 6.341 |

**Table 7** DIEHARD analysis results for the proposed audio scheme

| DIEHARD Test | p value | Ref. [10] |
|---|---|---|
| Birthday spacing | 0.59931 | 0.69363 |
| Overlapping 5-permutation | 0.20569 | 0.18709 |
| Binary rank (31×31) | 0.21470 | 0.32120 |
| Binary rank (32×32) | 0.86349 | 0.75071 |
| Binary rank (6×8) | 0.52476 | 0.45157 |
| Bitstream | 0.35047 | 0.48438 |
| Overlapping pairs sparse occupancy (OPSO) | 0.55630 | 0.55973 |
| Overlapping quadruples sparse occupancy (OQSO) | 0.61155 | 0.48439 |
| DNA | 0.57903 | 0.58096 |
| Count the 1's on a stream of bytes | 0.80632 | 0.45073 |
| Count the 1's for specific bytes | 0.96421 | 0.50060 |
| Parking lot | 0.96340 | 0.56747 |
| Minimum distance | 0.31547 | 0.21352 |
| Squeeze | 0.36987 | 0.54176 |
| 3D spheres | 0.25891 | 0.02040 |
| Overlapping sums | 0.78506 | 0.58811 |
| Craps | 0.89960 | 0.83608 |
| Runs up | 0.23257 | 0.31614 |
| Runs down | 0.25891 | 0.24235 |

### 5.12 ENT test

This statistical measure is utilized to check the randomness of the data. The test includes Entropy, Optimum compression, Chi-square distribution, Arithmetic mean value, Monte Carlo pie estimation, and Serial correlation coefficient. The results are presented in Table 8. The ENT results are also compared with the results published in Ref. [10]. The comparative analysis shows that our results are highly random and robust against cryptographic attacks.

## 6 Conclusion

A new design of audio signal encryption is evaluated in this article. Gensio-Tesi map-based substitution permutation network is utilized to accomplish an ideal cryptographic algorithm. To examine the security of the algorithm proposed cryptographic algorithm is qualified through some standard analysis. The frequency, amplitude, spectrum, and spectrograms of the verified audio signals reveal the modifications in encrypted data as compared to the original data. Moreover, NIST test results show that the encrypted audio exhibits maximum randomness. The computed MSE and PSNR results demonstrate high levels of noise in the

**Table 8** ENT measures for offered encryption approach

| Test name | *p* value of the proposed scheme | Ref. [10] |
|---|---|---|
| Entropy | 7.9999 | 7.999999 |
| Optimum compression | Reduced 125,000,000 bytes by 0% | 0% |
| Chi-square distribution | 257.5869 | 253.43 |
| Arithmetic mean value | 129.4001 | 127.4968 |
| Monte Carlo pie estimation | 3.14171 | 3.14181006 |
| Serial correlation coefficient | −0.00001 | −0.000087 |

encrypted data, revealing the plain signal is shattered in the encryption procedure. Performed examination indicates the essential level of security against brute-force attacks. Contemplating the acquired outcomes through the security examination, we can determine that the suggested scheme has crucial cryptographic security for encrypted audio signals. In the future, the proposed encryption structure can also be extended to image and video encryption strategies. The number of bits in the proposed substitution box can be reduced to transform the system into lightweight encryption strategies.

**Data availability** The authors confirm that the data supporting the findings of this study are available within the article and its supplementary material. Raw data that support the finds of this study are available from the corresponding author, upon reasonable request.

## Declaration

**Conflict of interest** The authors have not any conflict of interest regarding the publication of this article.

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